

Required Datasets for the Updated UrbanVPRM Model:

- **Data required for script 1 ‘processLC_ISA_script1.r’:**
 - Shape file for the boundaries of your region of interest
 - Need to make this yourself in a separate script
 - Land cover data from MODIS: [LP DAAC - MCD12Q1](#)
 - Impervious surface area (ISA) data (combination of GMIS, ACI, SOLRIS and City of Toronto’s impermeable surface datasets)
 - Need to pre-process, see supplementary material of <https://doi.org/10.1525/elementa.2023.00102> and ‘Toronto_permeability_plot_GMIS.r’
 - GMIS data: <https://doi.org/10.7927/H4P55KKF>
 - ACI data: [Annual Crop Inventory - Open Government Portal](#)
 - Southern Ontario Land Resource Information System (SOLRIS): <https://www.arcgis.com/home/item.html?id=0279f65b82314121b5b5ec93d76bc6ba>
 - City of Toronto Impermeable Surface: [Topographic Mapping – Impermeable Surface - City of Toronto Open Data Portal](#)
 - C3:C4 ratio derived from Annual Crop Inventory (ACI) data: [Annual Crop Inventory - Open Government Portal](#)
 - Needs to be pre-processed using ‘ACI_C3C4_fraction.r’ code
- **Data required for script 2 ‘processEVI_LSWI_script2.r’:**
 - Processes land cover data from script 1
 - MODIS MOD09A1 reflectance data:
 - Can be downloaded as TIFF files from [AppEEARS](#)
 - Need to specify boundaries of region of interest before downloading
 - Must include the red (b01), NIR (b02), blue (b03), and SWIR (b07) surface reflectance bands plus the quality flag layer (sur_refl_qc_500m), and day of the year
 - You must also include the latest file (doy 361) from the previous year and two files from the next year (doy 1 and 9) so that the code can properly interpolate missing EVI data at the start and end of the year
- **Data required for script 3 ‘processRAP_script3.r’:**
 - Rapid Refresh product (RAP) weather data: [Rapid Refresh/Rapid Update Cycle | National Centers for Environmental Information \(NCEI\)](#)
 - Download the 13km product but if there were any missing dates/times use the 20km data
 - Need to pre-process to turn RAP data from a grib format to a tiff
 - Can be done using ‘grb_to_tiff.sh’ script
 - This takes quite a while so budget time for this!!
 - Keep at least one .grib file to get the extent
 - **If there are still missing dates/times** use ERA5 (2m) air temperature data to fill in gaps: [ERA5-Land hourly data from 1950 to present](#)
- **Data required for pre-processing GOES data ‘pre_process_GOES.r’:**
 - GOES hourly surface solar irradiance: [GOES-East Surface Solar Irradiance | OSI SAF](#)
 - A csv file containing the following columns: year, month (mon), day of month (day), day of week (dow), hour of day (hour), hour of year (chr), day of year (cday), date (in format YYYYMMDD), datetime (in format YYYYMMDDHH). For the whole year (see **times2018.csv**). e.g.:

year	mon	day	dow	hour	chr	cday	date	datetime
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2019	1	1	6	0	1	1	20190101	2019010100
2019	1	1	6	1	2	1	20190101	2019010101
2019	1	1	6	2	3	1	20190101	2019010102
2019	1	1	6	3	4	1	20190101	2019010103
2019	1	1	6	4	5	1	20190101	2019010104
2019	1	1	6	5	6	1	20190101	2019010105
...	...	...	...	...	...	...	...	...
2019	12	31	6	20	8757	365	20191231	2019123120
2019	12	31	6	21	8758	365	20191231	2019123121
2019	12	31	6	22	8759	365	20191231	2019123122
2019	12	31	6	23	8760	365	20191231	2019123123

- **Data required for script 4 ‘processGOES_script4.r’:**
 - pre-processed GOES data from ‘pre_process_GOES.r’
 - csv file with date time data (see above)
 - Processed land cover data from ‘processLC_ISA_script1.r’
 - Processed RAP data from ‘processRAP_script3.r’

- **Data required for script 5 ‘runVPRM_script5.r’:**
 - Processed land cover, impervious surface, and C4:C3 ratio data from ‘processLC_ISA_script1.r’
 - .tiff file containing the percentage of water in each pixel. Can be made using ‘SOLRIS_ACI_water_plot.r’
 - Water data from the Southern Ontario Land Resource Information System (SOLRIS): <https://www.arcgis.com/home/item.html?id=0279f65b82314121b5b5ec93d76bc6ba>
 - Water data from the Annual Crop Inventory (ACI): [Annual Crop Inventory - Open Government Portal](#)
 - MODIS green-up and dormancy dates for each pixel. Can be made using ‘MODIS_greenup_dormancy.r’
 - Processed land cover data from ‘processLC_ISA_script1.r’
 - MODIS phenology data (MCD12Q2) greenup 1 & 2, dormancy 1 & 2, and quality flags (for each): can be downloaded as TIFF files from [AppEEARS](#)
 - Need to specify boundaries of region of interest before downloading
 - Need multiple years of data to fill in missing/unrealistic start/end dates
 - Processed EVI and LSWI data from ‘processEVI_LSWI_script2.r’
 - **Note:** need to identify a pixel in your domain that is fully forested (outside of the city) for use as a reference pixel
 - Processed RAP and GOES data from ‘processGOES_script4.r’